

Ismail

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Portfolio



USER ENGAGEMENT ANALYSIS

SQL(BigQuery)

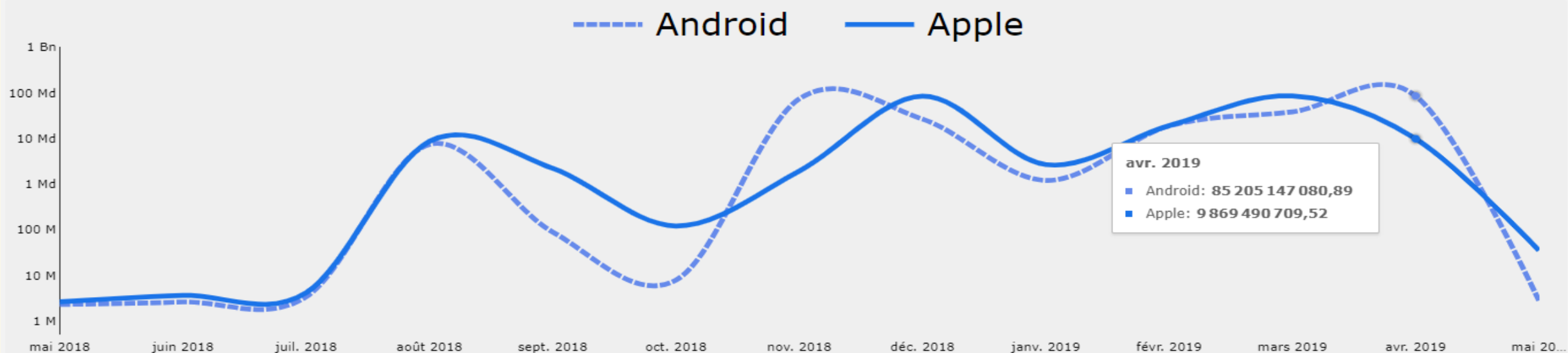
Looker Studio

Revolut



```
select
nbd.device,
date(date_trunc(created_date, month)) as month,
round(sum(amount_usd),2) as value,
count(transaction_id) as transactions
from `neo_bank.transactions` nbt
join `neo_bank.devices` nbd on nbt.user_id = nbd.user_id
group by 1,2
order by 1 asc;
```

Exchanged value



> Both device users follow similar trends, but the variations among Android users are more pronounced.

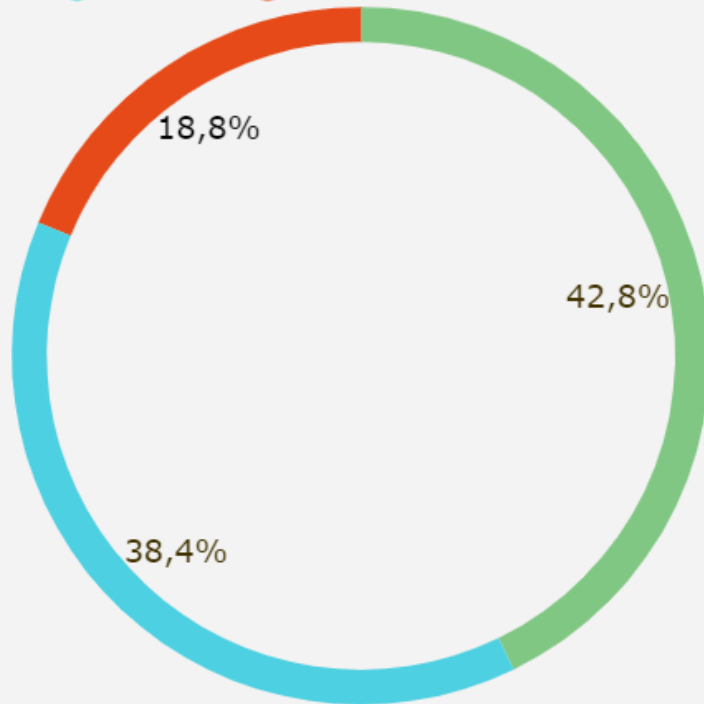


-- Channel sent rate

```
select channel,  
round(sum(case when status = 'SENT' then 1 end)  
/count(status)*100,2) as nb_sent  
from `neo_bank.notifications`  
group by 1  
order by 2 desc;
```

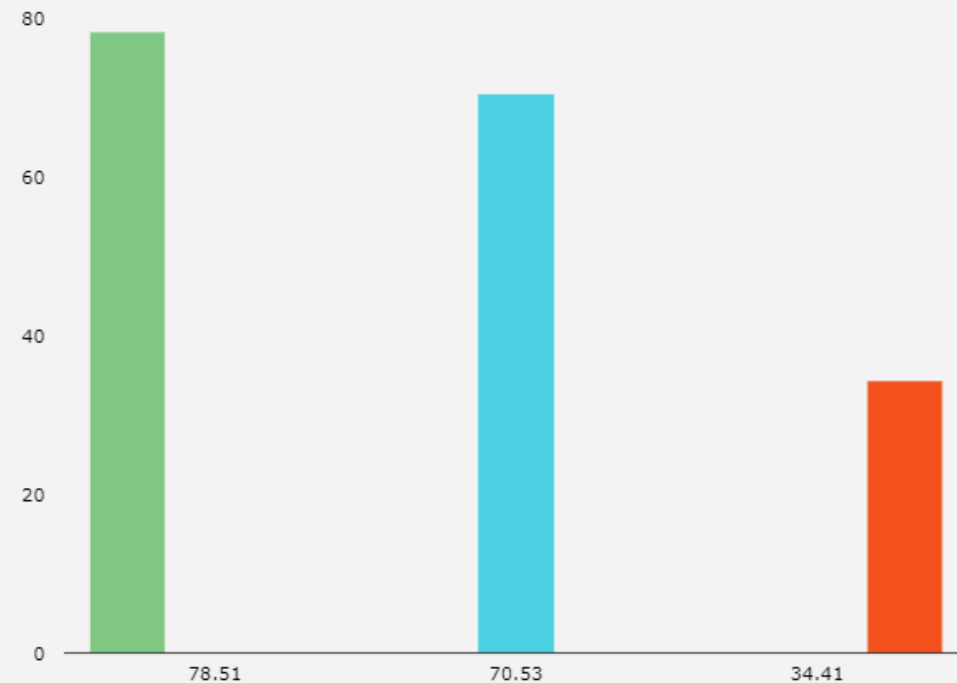
Channel utilization

PUSH EMAIL SMS



Successfully sent

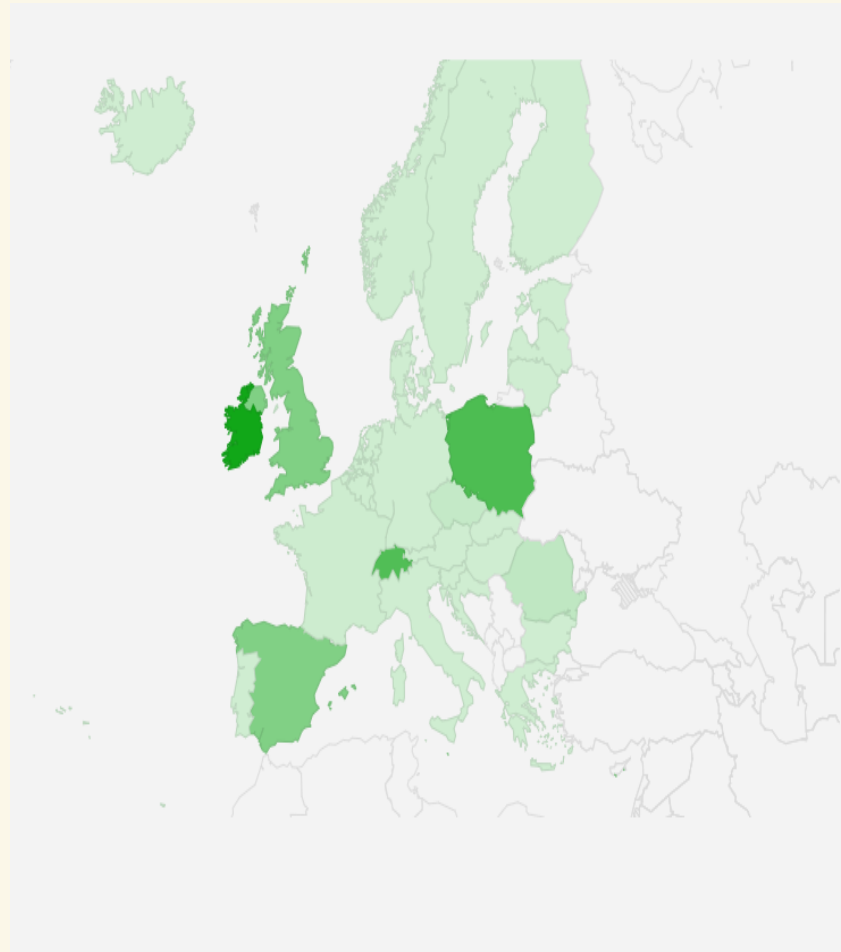
PUSH EMAIL SMS



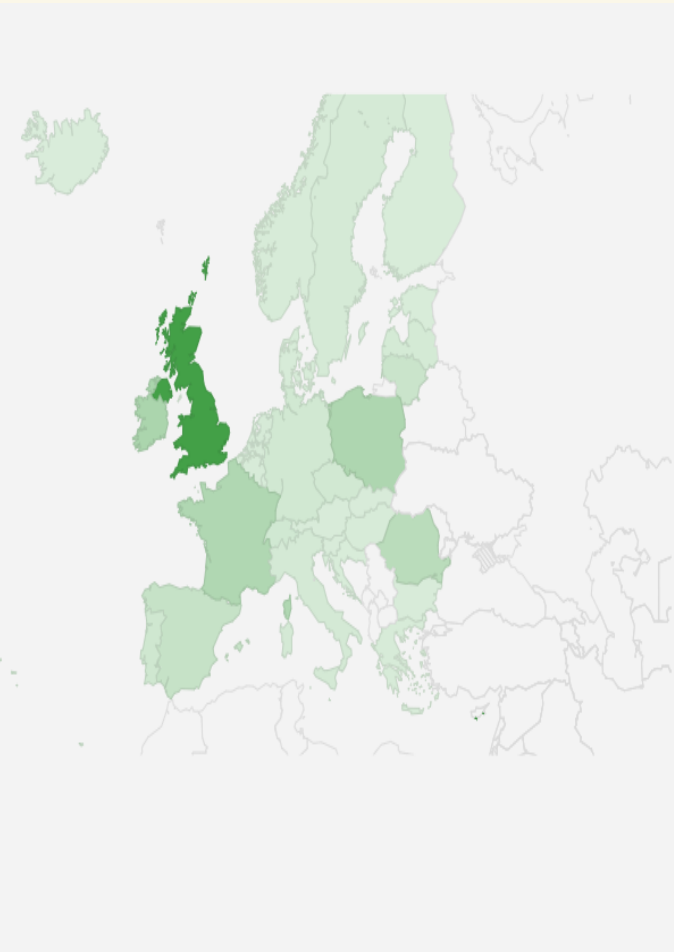
WHERE AND HOW MONEY IS SPENT?

- *EUR is the most utilized currency, representing most of the exchanged value.*
- *CHF (Switzerland) accounts for 17.76% of the exchanged value, while it only represents 1.45% of transactions.*
- *Despite GBP constitutes 11.1% of exchanged value, while it accounts for 34,32% of transactions.*

Value



Transactions



Few SQL queries

```
-- Currency usage value and transactions

with
p_trans as (select count (transaction_id) as tot_trans from `neo_bank.transactions`),
p_am as (select sum(amount_usd) as tot_am from `neo_bank.transactions`)
select
transactions_currency,
round(sum(amount_usd)/(select tot_am from p_am)*100,2) as pct_amount,
round(count(transaction_id)/(select tot_trans from p_trans)*100,2) as pct_trans
from `neo_bank.transactions`
group by 1
having pct_amount > 0.0
order by 2 desc;
```



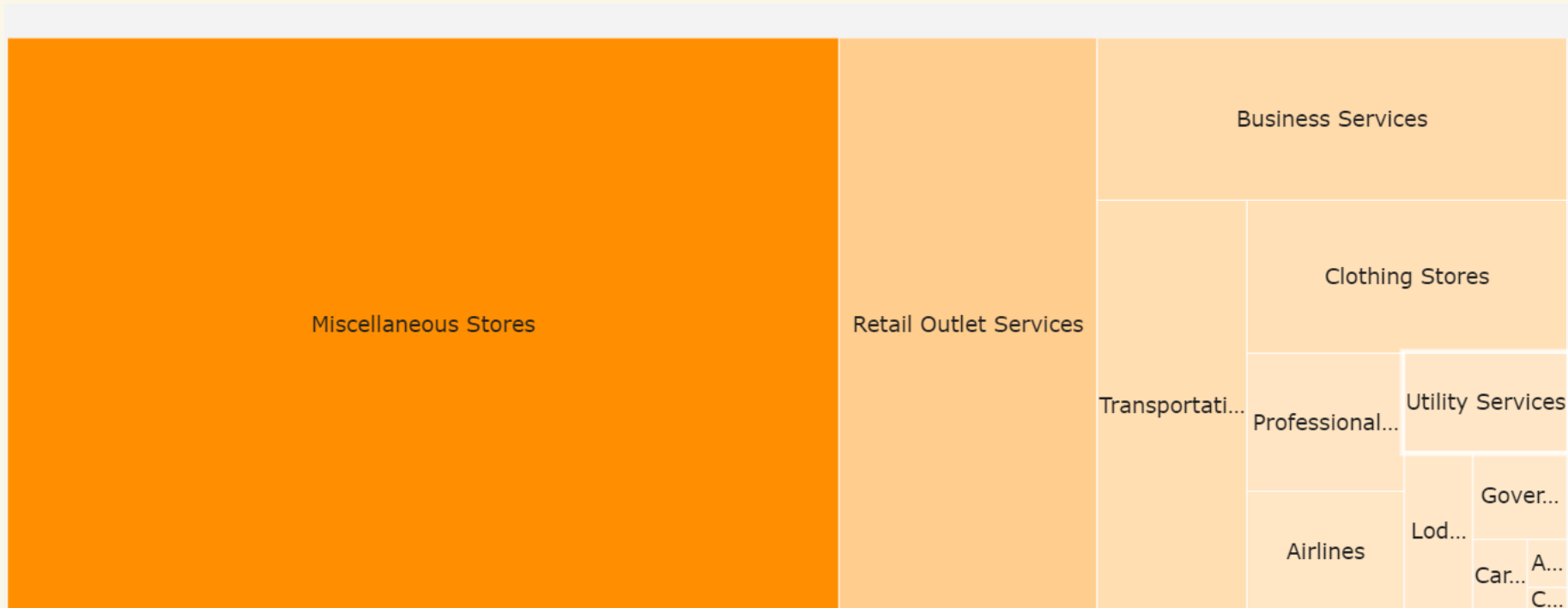
Few SQL queries

```
with
p_am as (select sum(amount_usd) as tot_am from `neo_bank.transactions`)
select
transactions_currency as currency,
round(sum(amount_usd)/(select tot_am from p_am)*100,2) as pct_amount
from `neo_bank.transactions`
group by 1
having pct_amount > 0.0
order by 2 desc;

with
p_trans as (select count (transaction_id) as tot_trans from `neo_bank.transactions`)
select
transactions_currency,
round(count(transaction_id)/(select tot_trans from p_trans)*100,2) as pct_trans
from `neo_bank.transactions`
group by 1
order by 2 desc;
```



Looker Studio Dashboard



> The majority of spending occurs at small to medium-sized local stores.

This highlights the importance of ensuring Revolut payments remain accessible for these types of businesses.




```

-- mcc category value distrib

with
total_amount as (select sum(amount_usd) as t_a from `neo_bank.transactions` where ea_merchant_mcc is not null)
select
distinct(case
when ea_merchant_mcc >= 0001 and ea_merchant_mcc <= 1499 then 'Agricultural Services'
when ea_merchant_mcc >= 1500 and ea_merchant_mcc <= 2999 then 'Contracted Services'
when ea_merchant_mcc >= 3000 and ea_merchant_mcc <= 3299 then 'Airlines'
when ea_merchant_mcc >= 3300 and ea_merchant_mcc <= 3499 then 'Car Rental'
when ea_merchant_mcc >= 3500 and ea_merchant_mcc <= 3999 then 'Lodging'
when ea_merchant_mcc >= 4000 and ea_merchant_mcc <= 4799 then 'Transportation Services'
when ea_merchant_mcc >= 4800 and ea_merchant_mcc <= 4999 then 'Utility Services'
when ea_merchant_mcc >= 5000 and ea_merchant_mcc <= 5599 then 'Retail Outlet Services'
when ea_merchant_mcc >= 5600 and ea_merchant_mcc <= 5699 then 'Clothing Stores'
when ea_merchant_mcc >= 5700 and ea_merchant_mcc <= 7299 then 'Miscellaneous Stores'
when ea_merchant_mcc >= 7300 and ea_merchant_mcc <= 7999 then 'Business Services'
when ea_merchant_mcc >= 8000 and ea_merchant_mcc <= 8999 then 'Professional Services'
when ea_merchant_mcc >= 9000 and ea_merchant_mcc <= 9999 then 'Government Services'
end) as mcc,
round(sum(amount_usd)/(select t_a from total_amount)*100,2) as p_value
from `neo_bank.transactions`
where ea_merchant_mcc is not null
group by 1
order by 2 desc;

```

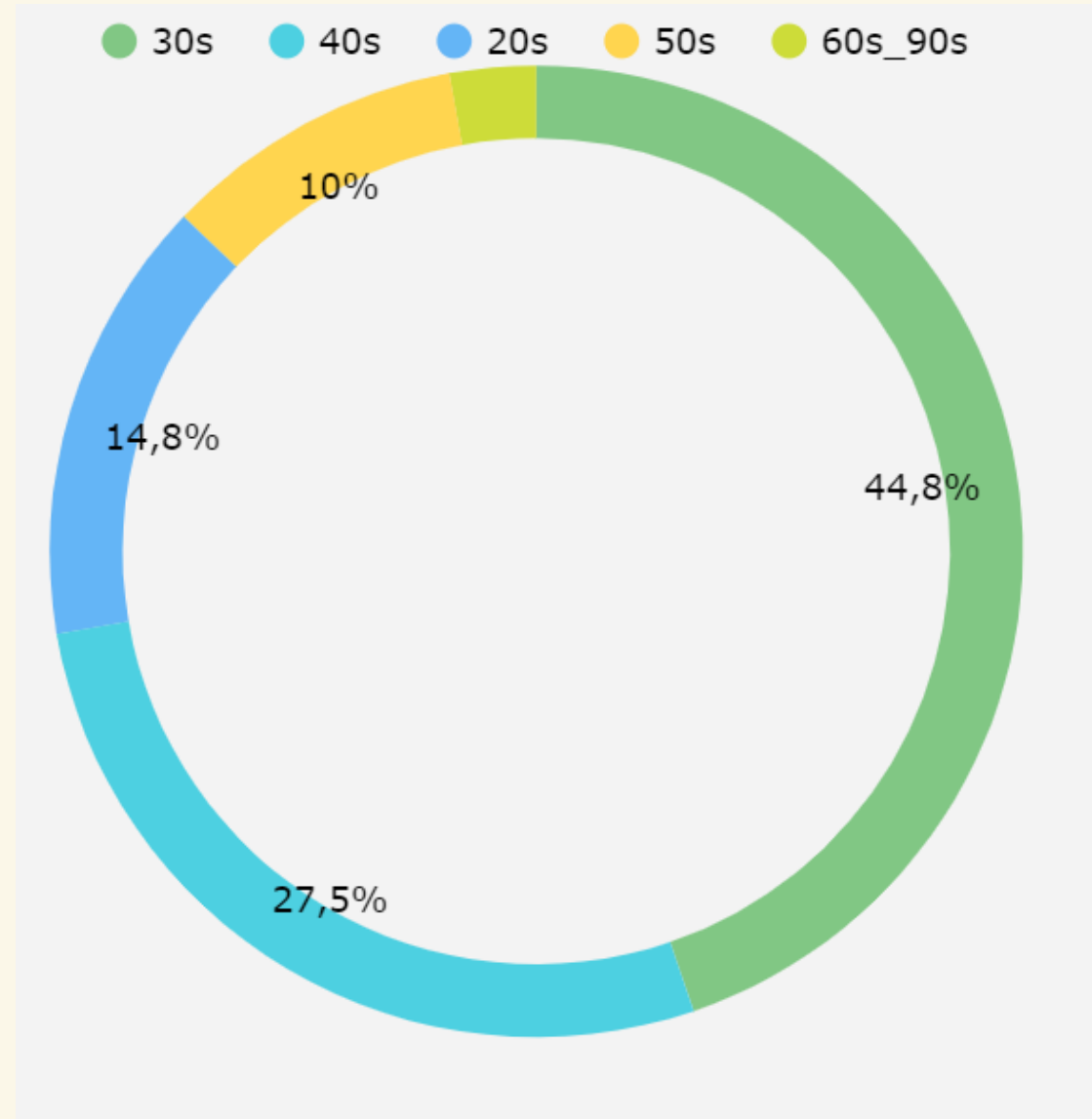


Crypto adoption

Users in their 30s constitute the largest segment and account for the highest value of exchanged transactions.

Additionally, they are the most prominent group using cryptocurrency, followed by users in their 40s and then those in their 20s.

The 20s and 40s are competing for the second and third positions, with the 40s being the second largest age group, while the 20s represent the second highest value of exchanged transactions.

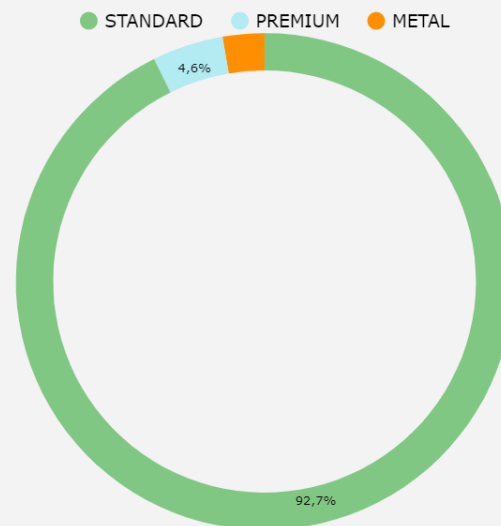


Potential subscription upgrades

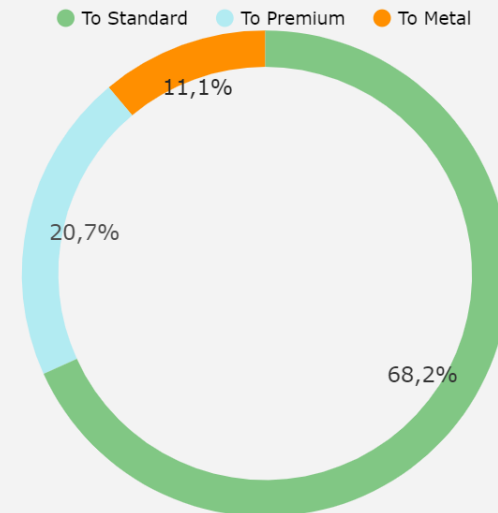
3.06% will potentially upgrade to upper subscription plan.

When users reach certain exchanging volumes and transactions quantity thresholds, tend to upgrade their plan.

Total users
18 766



Potential upgrades
576



```
-- Potential upgrade profiles
```

```
-- segmentation establishment
```

```
select
case
when nbu.plan in ('PREMIUM_FREE', 'METAL_FREE','PREMIUM_OFFER' ) then 'free'
when nbu.plan = 'PREMIUM' then 'premium'
when nbu.plan = 'METAL' then 'metal'
when nbu.plan = 'STANDARD' then 'standard'
end as plan,
round(sum(nbt.amount_usd)/count(nbu.user_id),2) as avg_value_exchanged,
round(count(transaction_id)/ count(distinct nbu.user_id),2) as nb_trans
from `neo_bank.transactions` nbt
left join `neo_bank.users` nbu on nbt.user_id = nbu.user_id
group by 1
order by 2 asc;
```

```
--> standard and premium users make less transactions than free and metal users but for bigger amounts on average
```

```
--> metal users make more transactions and bigger amount on average than free users
```

```
-- Potential upgrades
```

```
SELECT
nb_u.user_id,
nb_u.plan AS current_plan,
COUNT(nb_t.transaction_id) AS nb_transactions,
ROUND(SUM(nb_t.amount_usd), 2) AS total_value,
CASE
WHEN nb_u.plan in ('PREMIUM_FREE', 'METAL_FREE','PREMIUM_OFFER' ) AND SUM(nb_t.amount_usd) > 222.87 THEN 'Potential Upgrade to Metal'
WHEN nb_u.plan = 'METAL' AND SUM(nb_t.amount_usd) > 6253.43 THEN 'Potential Upgrade to Standard'
WHEN nb_u.plan = 'STANDARD' AND SUM(nb_t.amount_usd) > 179165.06 THEN 'Potential Upgrade to Premium'
ELSE 'Stable'
END AS upgrade_likelihood
FROM `neo_bank.transactions` nb_t
JOIN `neo_bank.users` nb_u ON nb_u.user_id = nb_t.user_id
GROUP BY nb_u.user_id, nb_u.plan;
```